Data Structures Assignment - Trees & Graphs

Binary Search Tree (BST)

A Binary Search Tree is a data structure where the left child is smaller than the root and the right child is greater than the root.

Why inorder traversal gives sorted order

In inorder traversal, nodes are visited in the order: Left subtree, Root, Right subtree. This ensures elements are printed in ascending order.

Time Complexity

Average case: O(log n). Worst case (skewed tree): O(n).

Graph Traversal

BFS (Breadth First Search)

BFS visits nodes level by level using a queue.

Use case: Finding shortest path in an unweighted graph.

DFS (Depth First Search)

DFS explores nodes deeply before backtracking using recursion or stack.

Use case: Cycle detection.

Observations

BST is efficient for searching and sorting. BFS ensures shortest traversal level-wise. DFS is useful for exploring deep paths.

Conclusion

This assignment helped in understanding tree operations and graph traversal techniques used in real-world applications like routing and databases.